

★ WEEK 15 · PART 2 · PRACTICE

Hedging and Risk, Worked by Hand

A Dynamic Hedge Ratio and a Breathing VaR

Compute h from five days; update volatility and VaR by hand

Part 2 at work, with the arithmetic shown. We compute a minimum-variance hedge ratio from five days, measure the variance it removes, update volatility with one shock, and turn it into a Value-at-Risk that breathes.

2P

PART 2 · APPLICATIONS

Two Desks, Computed by Hand

Case A: compute the hedge ratio h from five days, see how much risk it cuts, then update it. Case B: update volatility with one big return, then read off a VaR that a fixed limit would miss. Then a Claude Code lab.

What Part 2 Gave Us

Now we put them to work - with numbers

1

FFBS draws whole state paths

A two-block Gibbs sweep alternates states and parameters; FFBS samples a full path per sweep, giving a median with a credible band.

2

The hedge ratio becomes a fan

A drifting coefficient, inferred as a hidden state, turns one hedge number into a band. We first compute that number by hand, then update it.

3

TVP-SV lets volatility breathe

A stochastic-volatility state makes the noise level move. We approximate it by hand with one EWMA update, then read off the VaR.

Everything here is a calculator exercise: five days of hedging data, one volatility update, one VaR.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the key number by hand, step by step.
- Then update it as new data arrives.

What you should watch

- How much risk the hedge actually removes.
- How volatility jumps after one shock.
- How a fixed limit misses that jump.

The Dynamic Futures Hedge

A hedge ratio that must track a drifting optimum

A firm hedges a spot position with futures. The minimum-variance hedge ratio is $\text{Cov}(\text{spot}, \text{futures})$ over $\text{Var}(\text{futures})$. We compute it from five days, measure how much variance it removes, then update it when the next window disagrees.

The setup

- A spot position to protect with futures.
- The best hedge is the minimum-variance ratio.
- That ratio shifts as correlations move.

What we'll do by hand

- Compute h from five days.
- Measure the variance it removes.
- Update h for a new window.

The Minimum-Variance Hedge Ratio

And why it drifts

$$h_t^* = \frac{\text{Cov}(\Delta S, \Delta F)}{\text{Var}(\Delta F)} \text{ estimated as a drifting regression coefficient } \beta_t$$

What it is

- The covariance of spot and futures over the futures variance.
- The slope that minimizes hedged variance.
- Equivalently, a regression coefficient.

Why model it dynamically

- That covariance is not constant.
- So the optimal ratio drifts.
- A random-walk coefficient captures it.

The Data: Five Days

Compute the hedge ratio h by hand

Day	Futures dF (%)	Spot dS (%)
1	+1.0	+1.2
2	-0.8	-0.5
3	+1.5	+0.9
4	-1.2	-1.0
5	+0.6	+0.9

Goal: $h = \text{Cov}(\text{spot}, \text{futures}) / \text{Var}(\text{futures})$. We compute it in three lines on the next slide.

$h = \text{Cov} / \text{Var}$

Three lines with a calculator

$$\begin{aligned}\overline{\Delta F} &= 0.22, & \overline{\Delta S} &= 0.30 \\ \text{Cov}(\Delta S, \Delta F) &= \frac{1}{5} \sum (\Delta F - \overline{\Delta F})(\Delta S - \overline{\Delta S}) = \frac{4.36}{5} = 0.872 \\ \text{Var}(\Delta F) &= \frac{1}{5} \sum (\Delta F - \overline{\Delta F})^2 = \frac{5.448}{5} = 1.090 \\ h^* &= \frac{\text{Cov}}{\text{Var}} = \frac{0.872}{1.090} = \mathbf{0.80}\end{aligned}$$

The minimum-variance hedge is $h = 0.80$: short 0.8 futures for each unit of spot. Now, how much risk does that remove?

How Much Risk It Removes

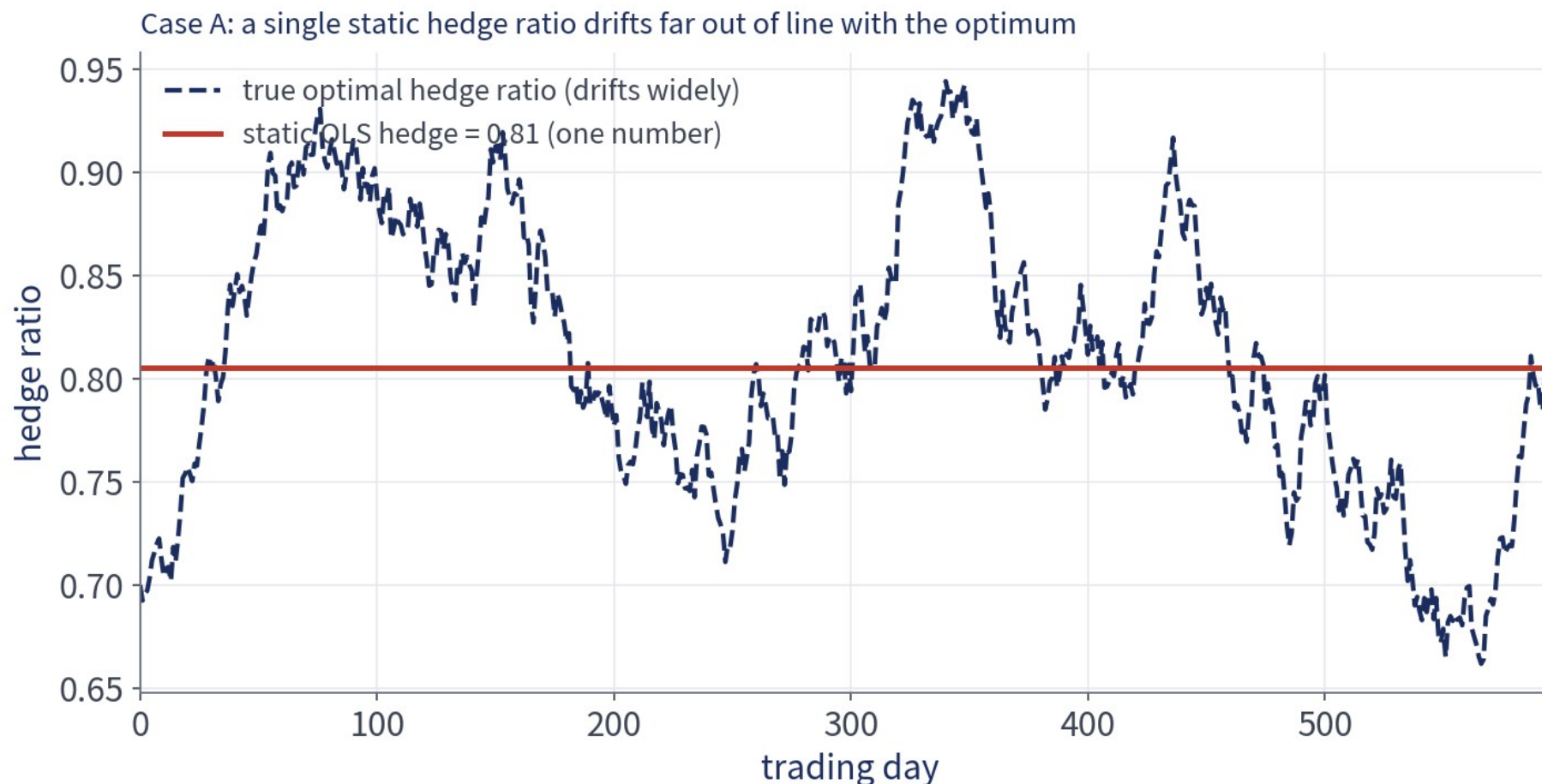
The whole point of a hedge

$$\begin{aligned}\text{unhedged: } \text{Var}(\Delta S) &= 0.77 \\ \text{hedged: } \text{Var}(\Delta S - 0.80 \Delta F) &= 0.074 \\ \text{risk cut} &= 1 - \frac{0.074}{0.77} \approx \mathbf{90\%} \text{ less variance}\end{aligned}$$

Hedging with $h = 0.80$ cuts the position's variance by about 90%. A wrong h would leave far more residual risk.

Static Hedge vs the Drifting Optimum

Compute that hedge ratio once over the whole sample and you get one flat number. But the true optimum drifts widely: the static hedge is right on average and wrong most of the time - over- and under-hedging



Update the Hedge Ratio

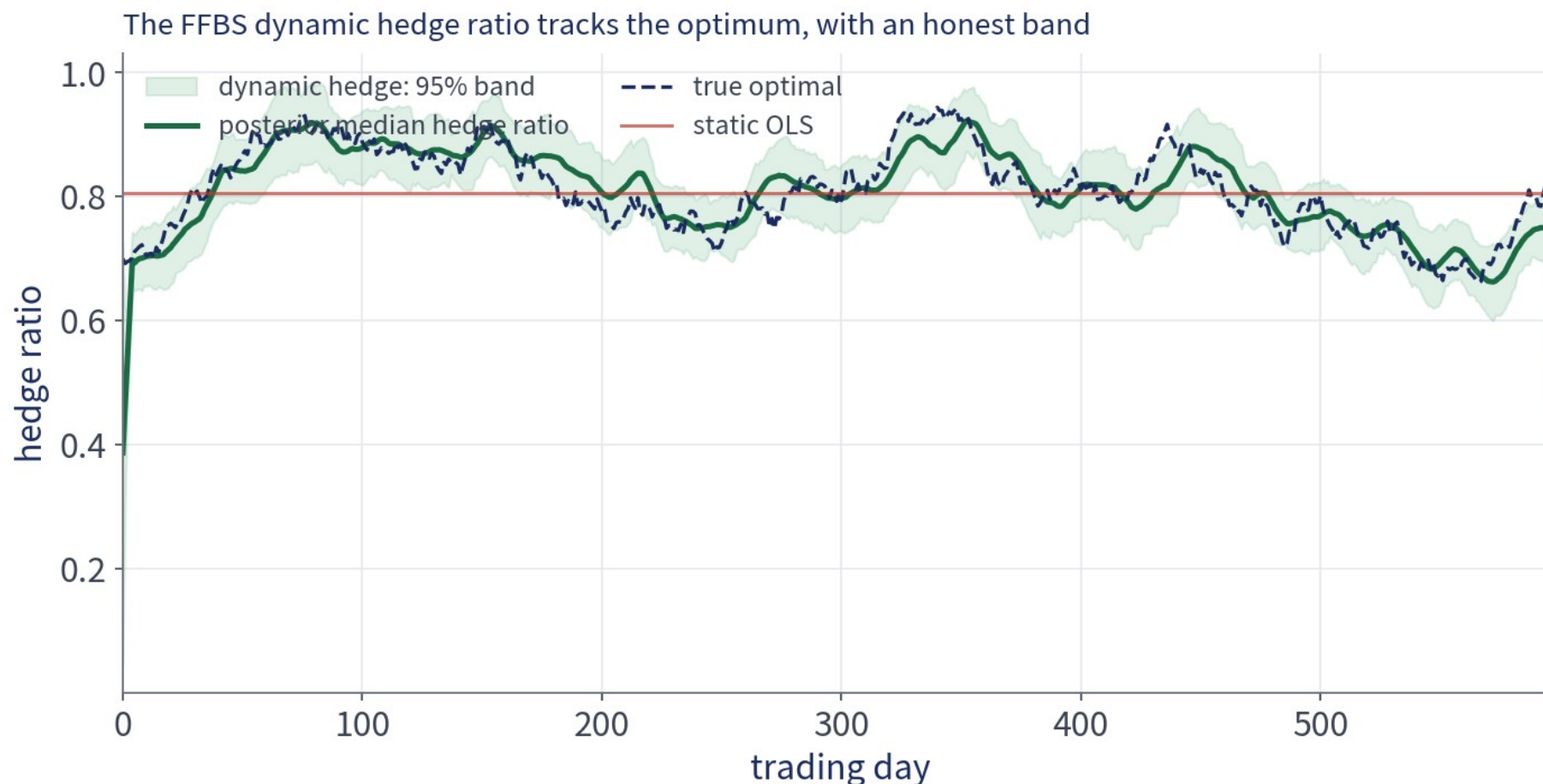
Prior + new window = posterior, by hand

$$\begin{aligned}\text{prior } h &\sim N(0.80, 0.15^2), & \text{new window } y &= 1.00 \pm 0.10 \\ \text{precision} &= \frac{1}{0.15^2} + \frac{1}{0.10^2} = 44.4 + 100 = 144.4 \\ \text{posterior mean} &= \frac{44.4(0.80) + 100(1.00)}{144.4} = \frac{135.6}{144.4} = \mathbf{0.94} \pm 0.083\end{aligned}$$

The updated hedge is a precision-weighted average: 0.94, pulled toward the sharper new window, with its own band of 0.083.

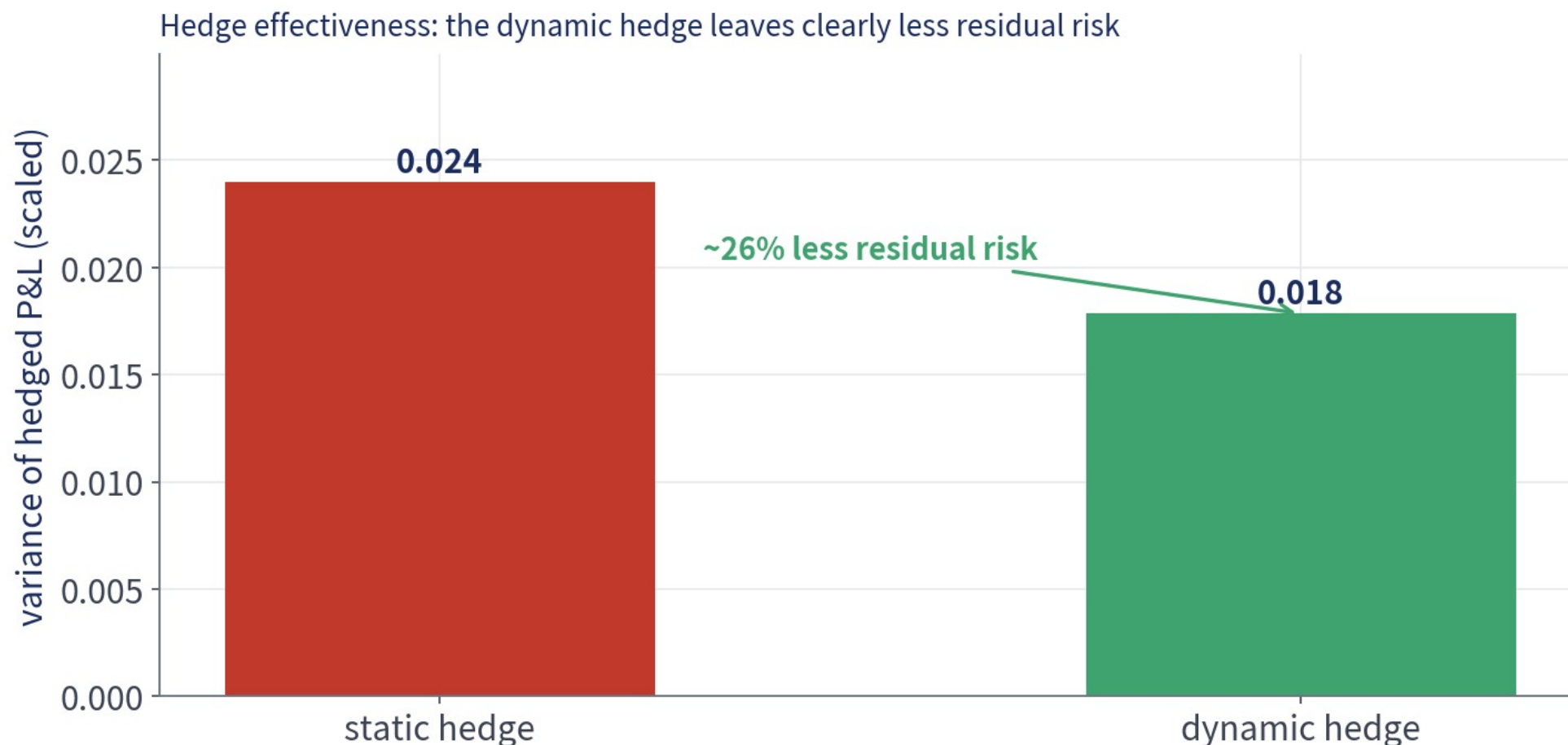
The FFBS Dynamic Hedge Ratio

Repeating that update every day gives a hedge ratio that tracks the drifting optimum and carries a credible band. The band widens where the relationship is noisy - an honest signal to hedge more cautiously



The Payoff: Less Residual Risk

Over the full sample, the same story as the hand calculation: because it stays aligned with the drifting optimum, the dynamic hedge leaves clearly less residual variance than the single static number.



What the Stale Hedge Ratio Costs

Basis risk you didn't sign up for

With the static ratio

- The hedge is mis-sized whenever the optimum moved.
- Residual variance leaks back into the position.
- You believe you are hedged more than you are.

With the dynamic ratio

- The hedge follows the optimum as it drifts.
- Residual risk stays low through the changes.
- The band flags when the hedge is less reliable.

Value-at-Risk That Breathes

A risk limit built on one volatility number

A risk desk sets a Value-at-Risk from a volatility estimate. We update volatility by hand with one big return using EWMA, watch it jump, then compute the VaR - and find a fixed limit that a real loss would breach while the breathing one holds.

The setup

- A daily VaR limit for a trading book.
- It scales directly with estimated volatility.
- One volatility number sets one fixed limit.

What we'll do by hand

- Update volatility after a shock (EWMA).
- Turn each volatility into a VaR.
- Find the loss that breaches the fixed limit.

A Value-at-Risk That Moves

Volatility as a hidden, breathing state

$$\text{VaR}_t = z_\alpha \sigma_t, \quad \sigma_t = e^{h_t/2}, \quad h_t = h_{t-1} + \nu_t \quad (\text{a hidden, moving level})$$

What it says

- VaR is a multiple of the current volatility.
- That volatility is the exponential of a log-vol state.
- The log-vol drifts as its own random walk.

Our hand version

- We approximate the moving vol with EWMA.
- It rises after a shock and decays in calm.
- Then VaR is just 1.645 times that vol.

Volatility After One Shock

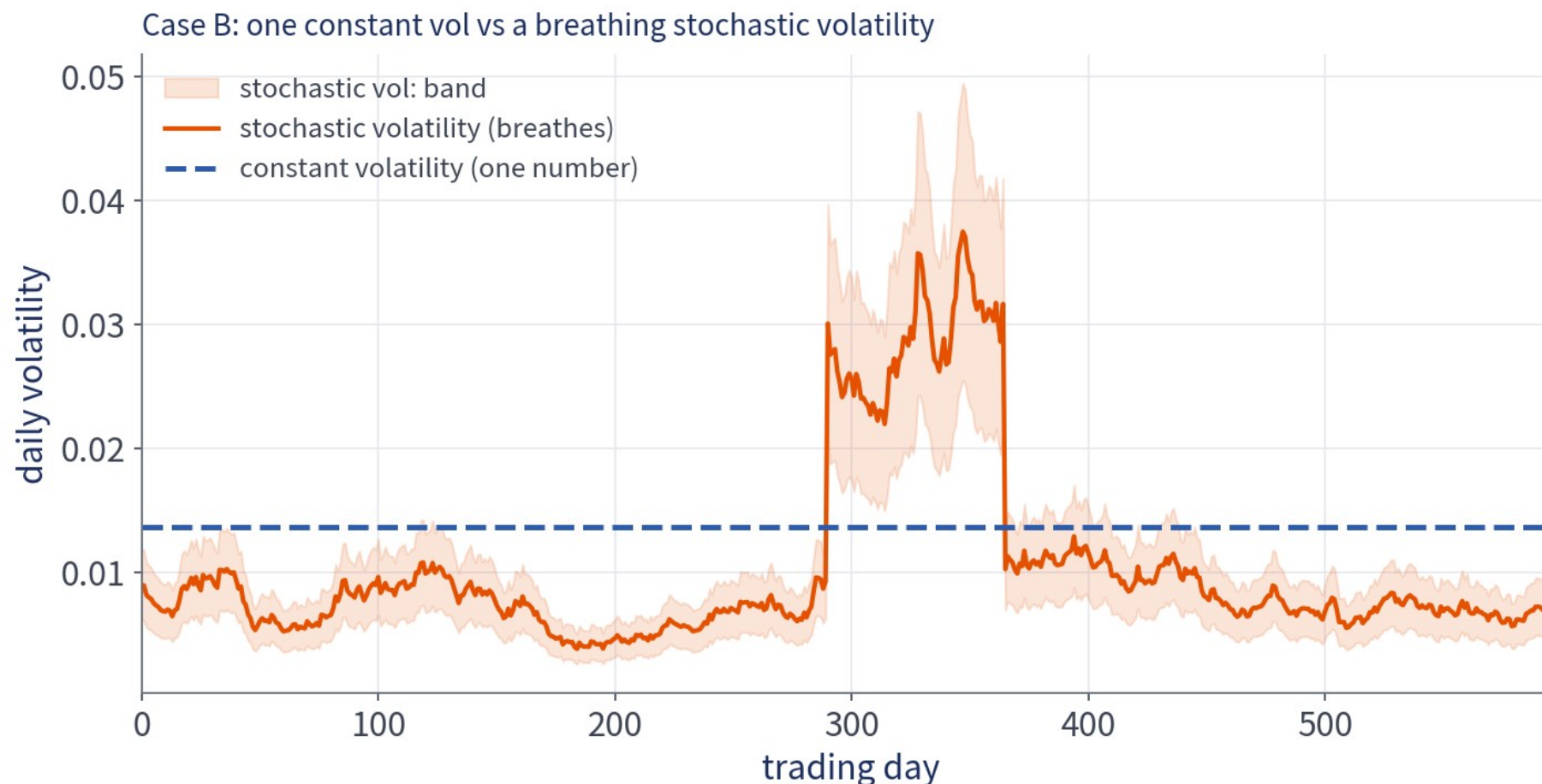
The EWMA update, by hand

$$\begin{aligned}\sigma_t^2 &= \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94 \\ &= 0.94(0.010)^2 + 0.06(-0.040)^2 \\ &= 0.000094 + 0.000096 = 0.000190 \\ \sigma_t &= \sqrt{0.000190} = \mathbf{1.38\%} \text{ (up from 1.00\%)}\end{aligned}$$

One 4% return drives volatility from 1.00% to 1.38% in a single step. This is how the estimate 'breathes' after stress.

Constant vs Stochastic Volatility

Do that update every day and volatility breathes: quiet for long stretches, then spiking in the stress window - exactly the jump we just computed by hand. A single constant volatility is one flat line through it all



VaR, and a Breach

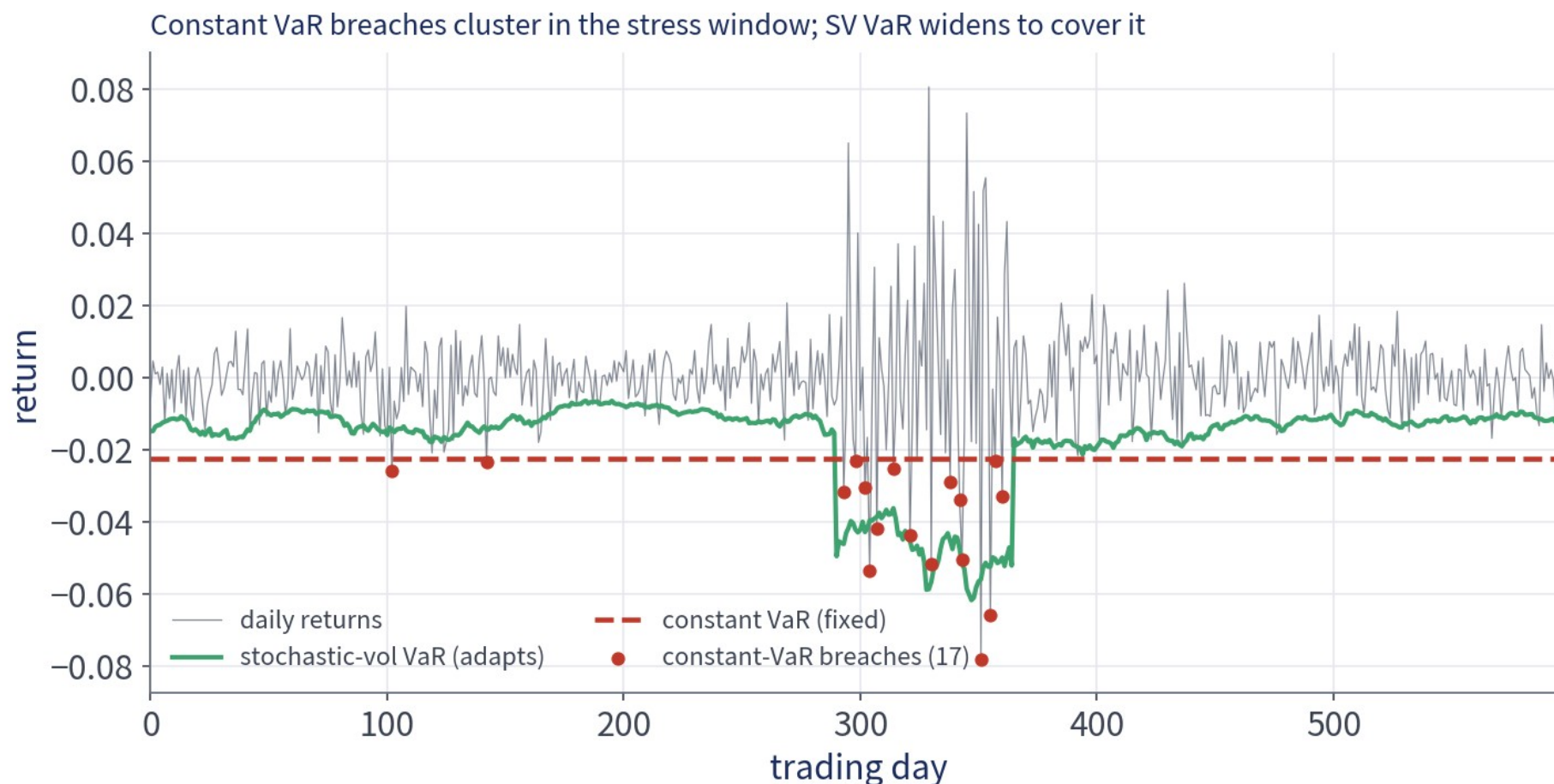
Turn volatility into a limit

$$\begin{aligned} \text{VaR}_{95\%} &= 1.645 \sigma \\ \text{calm: } 1.645(1.00\%) &= 1.65\%, & \text{after shock: } 1.645(1.38\%) &= 2.27\% \\ \text{constant (avg } 1.2\%) : 1.645(1.2\%) &= 1.97\% \\ \text{a } -2.1\% \text{ day: } \underbrace{2.1\% > 1.97\%}_{\text{breaches constant}}, & \underbrace{2.1\% < 2.27\%}_{\text{covered by SV}} \end{aligned}$$

A 2.1% loss breaches the fixed 1.97% limit but sits inside the post-shock 2.27% limit. The breathing VaR saw the storm coming; the fixed one didn't.

Constant VaR Breaks in Clusters

Over a full path, the fixed VaR is breached again and again in the stress window - precisely when it matters. The stochastic-vol VaR widens to cover the storm, so its breaches stay rare and spread out.



The Quant-Quake Echo

When a stale risk number turns procyclical

How it snowballs

- A fixed limit is fine until volatility jumps.
- Then clustered breaches force sudden deleveraging.
- Forced selling raises volatility further.

The honest alternative

- A breathing limit tightens gradually as vol rises.
- Adjustments are smaller and earlier.
- Less of the abrupt, cascading unwind.

The Common Pattern

Two desks, one moving quantity

Both cases, now computed by hand, replace a frozen number with a state that moves. The hedge ratio drifts; the volatility breathes. In each, updating the estimate - a weighted average, an EWMA step - resizes the decision as the market changes.

The trap

- A quantity that should move, held fixed.
- A hedge or a limit pinned to that stale value.
- A failure that arrives at the worst time.

The fix

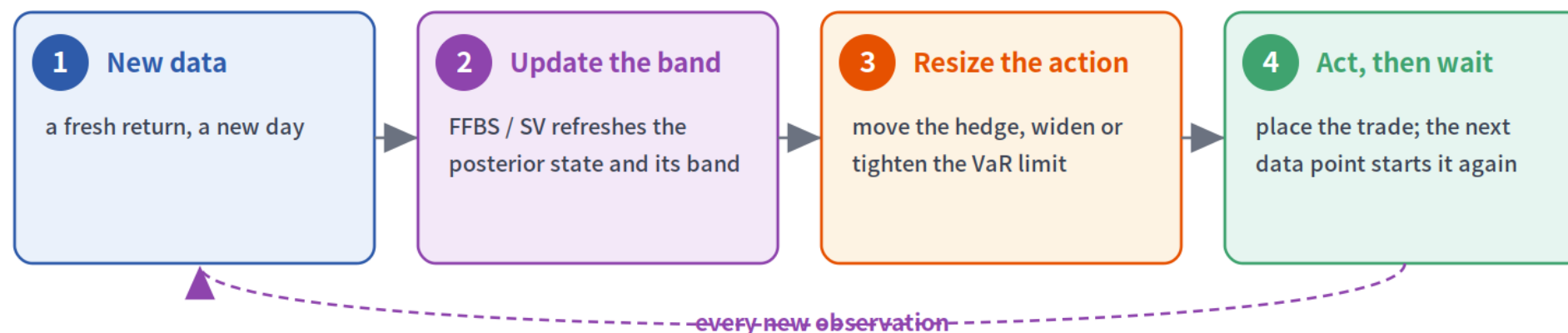
- Update the quantity as data arrives.
- Carry a band around it.
- Resize the hedge or limit as it moves.

A Decision That Moves With the Market

New data updates the estimate and its band - a weighted average for the hedge, an EWMA step for the vol; the band resizes the hedge or the risk limit; the next observation starts the loop again. The band is the input, not a report

A Decision That Moves With the Market

New data updates the posterior band; the band resizes the hedge or the risk limit. Then repeat.



Static once, or adaptive always

Case A resizes the hedge as the hedge ratio drifts; Case B resizes the VaR limit as volatility breathes. Both close the same loop: the posterior band is not a report - it is the input to the next decision.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to full time series.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - hedge:

- * take the 5-day `dF`, `dS` table; compute `Cov`, `Var`, and `h`, printing each number;
- * compute `Var(dS)` and `Var(dS - h*dF)`; report the percent risk reduction;
- * do the normal-normal update of `h` by hand-formula.

CASE B - VaR:

- * apply one EWMA update: $0.94 \cdot \text{sig}^2 + 0.06 \cdot r^2$;
- * print the new volatility;
- * compute calm, shock, and constant VaR at 1.645;
- * show which VaR a 2.1% loss breaches.

Match every printed number to the slides.

Check against the slides

- $h = 0.80$; risk cut ~90%.
- Updated $h = 0.94 \pm 0.083$.
- Vol 1.00% $\rightarrow$ 1.38% after shock.
- 2.1% loss breaches 1.97%, not 2.27%.

Deliverable: `part2_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get the hedge ratio from the five-day table?

$\text{Cov}(\text{spot}, \text{futures}) / \text{Var}(\text{futures})$

Q2 In one EWMA step, what makes volatility jump?

The large squared return term

Q3 Why does a 2.1% loss breach the fixed VaR but not the SV VaR?

The SV VaR widened after the shock

All three come from the hedge calc, the EWMA update, and the two VaR numbers.

Two Calculations to Remember

Part 2 applications in one screen

1

A hedge ratio is Cov/Var - and it cuts real risk

We computed $h = 0.80$ from five days, which removed about 90% of the position's variance. A static number drifts out of line; updating it (to 0.94) keeps the hedge aligned as the market changes.

2

Volatility breathes, and VaR must follow

One EWMA step took volatility from 1.00% to 1.38%. A 2.1% loss breaches the fixed 1.97% limit but not the post-shock 2.27% one - the breathing VaR saw the storm the fixed one missed.

3

The update is the input, not the output

In both desks, each update - a weighted average, an EWMA step - feeds the next decision, resizing the hedge and the limit as the market moves. That is what makes the response adaptive.

Next - Part 3 practice: soft regimes and structural breaks, also worked by hand.